

# Supplementary material

## How much of the evidence for climate-driven dengue transmission survives the analyst?

### A multiverse analysis of 221 outbreaks in 33 countries

Amna Dilber

Section numbers referenced from the main text. Every figure quoted here is produced by a numbered pipeline step and stored in `results/tables/`; the test suite checks the main text and this document against those tables together.

## S1 How far the decomposition depends on the factor list

More than the table above admits, and the honest version is worth more than the comfortable one. The three designs are nested — all contain the observation model — so agreement between them shows the answer survives adding cells to one factor set, not that it survives choosing another. We therefore recomputed it dropping each factor in turn, and over all 63 non-empty subsets of the six.

Table S1: The within-outbreak share under alternative factor lists. The result is load-bearing on the observation model and does not survive an adversarial choice of factor list, which is stated here rather than left to be found.

| Factor list                                      | Within-outbreak share |
|--------------------------------------------------|-----------------------|
| All six factors                                  | 76.5%                 |
| Dropping the fixed parameters                    | 76.7%                 |
| Dropping the model structure                     | 73.6%                 |
| Dropping the rainfall lag                        | 73.0%                 |
| Dropping the temperature form                    | 71.4%                 |
| Dropping the fraction of series fitted           | 70.5%                 |
| <b>Dropping the observation model</b>            | <b>55.2%</b>          |
| Median over all 63 subsets                       | 54.5%                 |
| Best case for a sceptic (fixed parameters alone) | 2.3%                  |

Removing any factor except the observation model costs at most six points. Removing the observation model costs twenty-one, and the analytical share is then 55.2% — still a majority, but the claim is plainly load-bearing on that one choice. A reader who regards the observation model as fixed by good practice rather than chosen should read 55%, and one who regards only the entomological parameters as genuinely free should read 2.3%.

We think the first two positions are the defensible ones, and the evidence for that is in Section 3: the published review finds 18.3% of models using Poisson regression and 18.3% linear regression, with the negative binomial not appearing as a category. The observation model is not fixed by practice, because practice has not fixed it.

One feature of the design was said to push the other way, and the claim was never tested. Every combination within an outbreak is warm-started from a single anchor fit and shares that fit's estimate of the dispersion, so the 144 analyses are arguably more alike than 144 independent

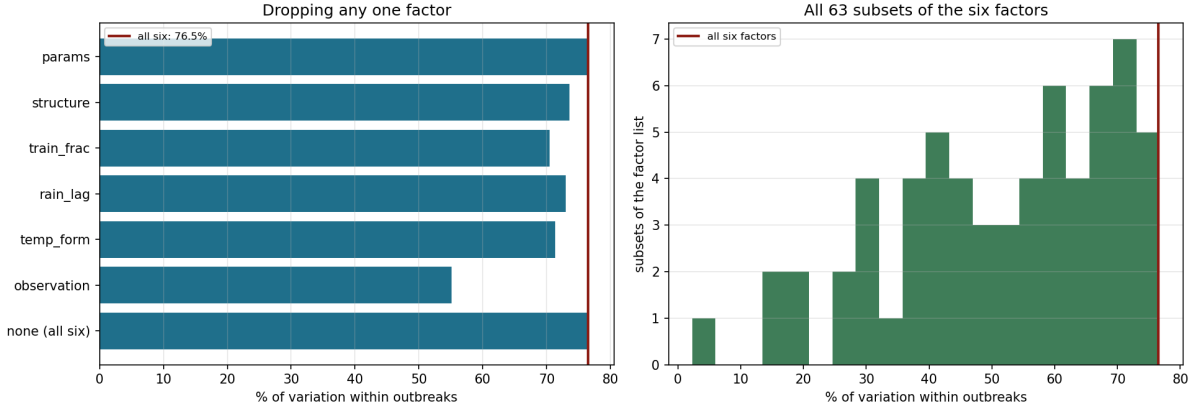


Figure S1: How far the headline depends on the factor list. Left: the within-outbreak share when each factor is dropped in turn. Right: the same quantity over all 63 non-empty subsets of the six factors. The claim is load-bearing on the observation model and does not survive an adversarial choice of factor list, which is why both panels are shown rather than the first alone.

analysts would be, and earlier drafts concluded that the share reported here is if anything an underestimate. The optimiser check of Section S5 refits seven outbreaks from ten cold starts with no warm start, which measures the direction rather than arguing it: pairwise disagreement is **33.4%** as run and **32.9%** cold. Cold starts give slightly *less* disagreement, not more. Seven outbreaks settle nothing about the size, and the check varies the warm start but not the shared dispersion, so one of the two mechanisms remains untested. We no longer claim to know the sign.

This does not show that local context is irrelevant. It shows that context has been asked to carry more than it can.

## S2 The published moderator does not reproduce here

The specific contextual explanation on offer is that the association is strongest where temperature varies most [1]. We tested its closest analogues on our own data — six descriptions of each outbreak’s temperature regime against three outcomes: the estimated Brière exponent, the share of analyses endorsing climate forcing, and the disagreement between two of them. Eighteen correlations, so about one would reach  $p < 0.05$  by chance and reporting raw significance would itself be the practice this paper measures.

Seven reach  $p < 0.05$ , against about one expected by chance, so the temperature regime is not unrelated to what the analyses find. But every association is weak — no  $|\rho|$  exceeds 0.22 — and **one survives control of the false-discovery rate**: the estimated temperature exponent against the season’s temperature range,  $\rho = -0.22$  ( $q = 0.02$ ), whose sign is the wrong way round for the hypothesis, the exponent being *smaller* where temperature varies more. The endorsement rate moves in the expected direction but does not survive correction ( $\rho = +0.15$ ,  $q = 0.10$ ). Both facts belong in the same sentence: there is something there, and it is small.

We do not read this as refuting the published result, which was computed on a different quantity: they measured the strength of a statistical correlation between incidence and temperature, we the exponent of a mechanistic response. What it does establish is that the moderator is not recoverable from a design in which the analysis is held fixed — which is what it would have to be if it were a property of dengue rather than of the estimates that reached print.

A mechanistic refinement we expected to do better also failed. Transmission responds to temperature along a unimodal curve, so a season spent near the flat optimum should offer nothing to detect while one crossing the steep flank offers a great deal; the mean gradient

traversed should therefore predict what the data can support. It does not ( $\rho = +0.13$ ,  $q = 0.11$ ), and it is reported because a study about undisclosed analytical choices does not get to drop the measure it hoped would work.

### S3 The instability is not specific to AIC

AIC, BIC and the likelihood-ratio test are all thresholds on the same quantity, placed at different points, so all three follow from what was already computed:  $\text{AIC} = -2\log L + 2k$  recovers the log-likelihood, and  $\Delta k = 2$  in every comparison.

Table S2: Instability under each criterion, used conventionally. All are thresholds on  $\Delta\text{AIC}$ ; the second column states where each one falls.

| Criterion                                     | Equivalent to $\Delta\text{AIC} <$ | Unstable     | 95% CI |
|-----------------------------------------------|------------------------------------|--------------|--------|
| AIC, sign                                     | 0.00                               | 91.9%        | 88–96% |
| Likelihood-ratio test, $p < 0.05$             | −1.99                              | 95.0%        | 91–98% |
| BIC, sign (at the median 38 weeks)            | −3.28                              | 96.8%        | 95–99% |
| Likelihood-ratio test, $p < 0.01$             | −5.21                              | 97.3%        | 95–99% |
| <b>AIC, abstain within <math>\pm 4</math></b> | —                                  | <b>39.8%</b> | 33–45% |
| BIC, abstain within $\pm 4$                   | —                                  | 90.5%        | 86–94% |

Strictness makes matters worse, monotonically. The significance test an epidemiologist would actually run is equivalent to a margin of about 2 and leaves 95% of outbreaks unstable. A margin of 4 on the BIC scale, which sounds like the same remedy, leaves 91%, because BIC’s additional penalty shifts the abstention band off centre — it abstains over  $(-7.3, +0.7)$  rather than  $(-4, +4)$ , so one side of the disagreement is still allowed to speak. The distinction that matters is not how strict a rule is but whether it abstains symmetrically about zero.

### S4 Why a small margin works and strictness does not

Two facts hold simultaneously and sound contradictory until separated.

**The choices move the evidence enormously.** The range of  $\Delta\text{AIC}$  across the 144 combinations of a single outbreak has a median of **804 AIC units** (quartiles 140 to 11,105). The median  $|\Delta\text{AIC}|$  being interpreted as evidence is 14.4. The analyst’s own choices move the evidence by roughly **fifty-six times** the size of the difference being read from it.

**Yet the dissent is marginal.** The margin needed to eliminate all disagreement within a window has a median of 4.00, and in **60.2%** of outbreaks every dissenting combination does so on evidence weaker than  $\Delta\text{AIC} = 4$ .

Here the six-factor design corrects a claim the smaller ones supported. On 48 combinations the margin needed was uncorrelated with how far the choices moved the evidence, and dissent looked like a pure property of comparisons that were weak to begin with. On 144 the correlation is  $\rho = +0.61$  ( $p = 4 \times 10^{-24}$ ): in the remaining 39.8% of outbreaks at least one dissenter is confident, and no band around zero can reach it. What stays true is that such dissenters are few — a median of 3 of the 100 analyses still speaking.

Both hold because most of that enormous variation stays on one side of zero: it changes how strong the evidence appears without changing which model it favours. In three outbreaks in five, everything that does cross zero crosses it weakly, and a band around zero reaches all of it. In the other two, something crosses decisively and no band will ever reach it — but when that happens the dissenter is nearly alone, a median of 3 of the 100 analyses still speaking.

That is why a margin of 4 removes almost all pairwise disagreement (29.8% to 4.0%) while leaving 39.8% of outbreaks with at least one dissenting cell. The two figures describe the same

fact from opposite ends and neither alone is honest: the surviving disagreement is common across outbreaks and rare within them.

It also explains why a stricter threshold cannot help. A stricter rule is still a rule with a boundary, and the confident dissent sits far from any boundary one might choose; moving the boundary only changes which of the weak comparisons are silenced.

We also tested and rejected a simpler explanation — that instability tracks the density of comparisons at the boundary. Scanning the threshold from  $-14$  to  $+2$  gives instability that runs backwards rather than forwards against the local density ( $\rho = -0.27$ ), so the hypothesis is not merely unsupported: it predicts the wrong sign.

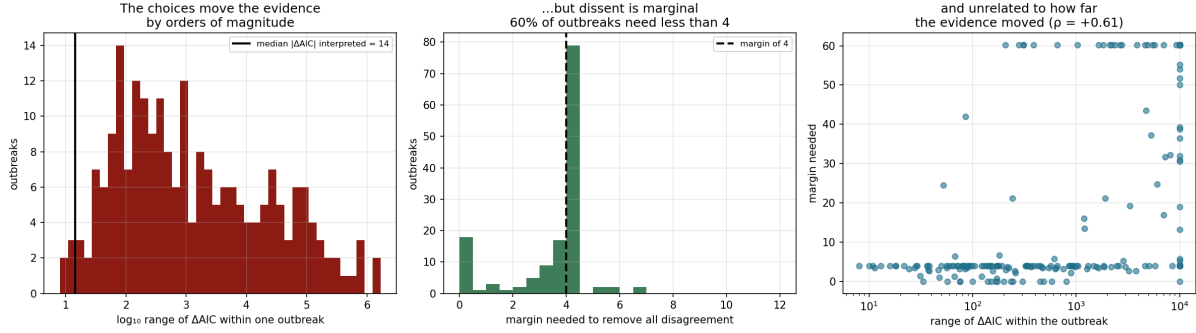


Figure S2: Left: the range of  $\Delta AIC$  induced within a single outbreak by the analysis choices, against the median difference interpreted as evidence. Centre: the margin needed to remove all disagreement within a window. Right: the two are unrelated.

## S5 Further checks

**Not a small-sample artefact.** By quartile of reported cases the unstable share is 94.6%, 90.9%, 89.1% and 92.7%; by quartile of series length, 89.5%, 93.2%, 96.1% and 88.9%. Neither ordering is monotone and no quartile falls below 88%. This is not a problem of thin data.

**Not an artefact of our own selection rule.** This study applied a peak-prominence threshold when choosing windows and must meet the standard it sets for others. Recomputing the headline on progressively stricter subsets gives 91.9%, 91.8% and 94.1% ( $n = 68$ ). Tightening our selection does not weaken the result.

**The climate grid cell matters, and it is measured rather than conceded.** A single representative point supplies the climate for a whole country or province, which earlier versions of this study named as a limitation and did not quantify. Forty outbreaks were therefore refitted under all 144 combinations at a second point one degree of latitude away — about 110 km, well inside the area any of these case series aggregates over. Across 5,664 paired comparisons the verdict changes in **12.0%**, with the endorsement probability essentially unmoved ( $0.739 \rightarrow 0.737$ ). That places the climate location between the covariate choices and the structural ones: above the transmission model, below the observation model, and nowhere near the top.

Read it as an upper bound rather than an estimate. The offset is mechanical and respects no terrain: in flat units the two series differ by a few tenths of a degree, which is the intended comparison, but for Costa Rica they differ by  $4.3^\circ\text{C}$  because the shift crosses high ground — a choice between two climates rather than between two defensible stations for one population. The alternative, hand-picking a second city per unit, would replace a reproducible rule with a gazetteer of our own choices, which is not a substitution this study is in a position to make quietly.

**Not optimiser noise.** The factorial used three multi-starts warm-started from an anchor fit, a compromise made for runtime. If three starts sometimes miss the optimum — and miss it

more often for the climate model, which has two extra parameters — then  $\Delta\text{AIC}$  would move for reasons unrelated to any analysis choice. Seven outbreaks were therefore refitted twice under the 48-combination design in place when the check was run: as run, and with ten cold restarts and no warm start. The thorough setting does find better optima, so the concern is not empty: it improves the climate fit in 6.1% of comparisons and the constant fit in 3.3%, and in one case by 1,656 AIC units. But it changes the verdict in **0.6%** of 329 paired comparisons, against 8.3% and 38.1% for the weakest and strongest analysis factors *in that same 48-combination design* — both bounds move when the factorial grows, so the comparison has to be made within one design — and every one of the seven outbreaks is unstable under both settings. A more thorough optimiser finds more, and what it finds does not move the conclusion. This check is small — seven outbreaks, run alongside the simulation study on the same machine — and is reported as a bound on optimiser noise rather than a precise estimate of it.

## S6 The Pakistani case study: what can be identified at all

The global study asks whether a conclusion is stable. It does not ask whether the underlying quantities are identifiable, which we examined in detail on Pakistani surveillance — a national series covering the complete 2013 wave and provincial series from 2021.

**Transmission is identified; catchment size often is not.** Profile likelihoods [2] bound  $R_0$  and the population at risk in the longer windows, but in a nineteen-week provincial window the population profile under the climate model is flat across a twelvefold range, while the constant model determines it to within a factor of two. Adding climate parameters to a short series destroys the identifiability of a parameter the simpler model recovers.

**Spatial aggregation depresses  $R_0$ .** Fitting districts separately and then fitting their sum gives different answers: a bias of  $-2.2\%$  in one province and  $-28.8\%$  in another, where the aggregate estimate of 1.14 falls below both constituent districts, 1.25 and 1.96. Component epidemics peak at different times, and a homogeneously mixing model can only read the broader sum as slower transmission.

**An independent check confirms it.** Bounding  $R_0$  from the observed early growth rate — using no compartmental fit, and reporting both the fixed and exponentially distributed generation-interval bounds — gives 2.91–6.77 nationally against a fitted 1.37. Every fitted value falls below the corresponding lower bound. A homogeneous model ties growth rate to final size through a single  $R_0$ ; these epidemics rise faster than their eventual size permits, which is the same spatial heterogeneity seen from a second direction.

**And the explanation was tested, not asserted.** Spatial heterogeneity is the natural account of both observations, but naming a mechanism is not demonstrating it. A two-patch model was therefore fitted to the Khyber Pakhtunkhwa *aggregate* alone — given no information about which districts contributed, how many there were, or when each peaked. It beats the single-patch fit by  $\Delta\text{AIC} = -8.6$ , recovers transmission rates of 1.72 and 2.53 that bracket the two districts instead of falling below both, clears the growth-rate lower bound the single-patch fit violated, and infers a 46-day offset between its components. Three predictions of the heterogeneity account, each checkable and each met, from a model that was told only the sum.

Fitted transmission rates from aggregated surveillance should therefore be read as lower bounds, not as characteristic values.

## References

- [1] D. Kirk, S. Straus, M. L. Childs et al., *Temperature impacts on dengue incidence are non-linear and mediated by climatic and socioeconomic factors: a meta-analysis*, PLOS Climate **3**(3), e0000152 (2024).

- [2] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller and J. Timmer, *Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood*, Bioinformatics **25**(15), 1923–1929 (2009).